

Walk-Unbiased Doubly Robust Graph Propagation: Provably Fair, Transparent, Accountable and Private Recommendation from Exposure-Biased Logs

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Abstract

Graph-based recommenders propagate preferences along multi-hop walks of the logged user-item graph. This graph is filtered by the platform’s own exposure policy, so propagation amplifies exposure bias. Recent debiasing methods reweight the edges of this graph by inverse propensities or impute missing edges with doubly robust (DR) estimates. We show that these corrections are still biased once propagation spans more than one hop. A walk that revisits an edge squares that edge’s estimate, and the squared inverse propensity inflates exactly the rarely exposed edges that debiasing is meant to recover. We derive the exact bias and a lower bound that grows as the inverse of the propensity clip. Because outcomes are binary, the fix is simple: every repeated edge is replaced by its unbiased idempotent counterpart. The resulting operator, *DRUP* (Doubly Robust, walk-Unbiased Propagation), is exactly unbiased and edge-wise doubly robust for propagation over the counterfactual full-exposure graph, and it costs no more than the uncorrected operator. The same structure gives *DRUP* provable properties beyond accuracy: (i) variance, clipping-bias and concentration bounds; (ii) exposure invariance, a causal item-fairness guarantee that the expected score does not depend on interventions on an item’s exposure, whereas logged-graph propagation amplifies exposure and uncorrected DR propagation over-corrects it; (iii) scores that are affine in each user’s own interactions, which yields exact Shapley attributions and provably minimal counterfactual explanations; (iv) closed-form certificates against injected fake profiles; and (v) a single public item operator whose Gaussian release makes the whole recommender jointly differentially private. On Coat and KuaiRec, whose test sets are random or fully observed, *DRUP* needs no training. Its accuracy matches or exceeds trained MF, LightGCN, NAVIP and DR baselines, and every guarantee is verified empirically.

Keywords: causal recommendation; graph neural networks; doubly robust estimation; exposure bias; fairness; explainability; certified robustness; differential privacy.

1 Introduction

Graph-based recommender systems (GBRs) represent users and items as nodes and logged interactions as edges. They score a user-item pair by aggregating signals along multi-hop walks [4, 14]. The logged graph, however, is not the preference graph. An edge exists only if the platform *exposed* the item and the user then responded. A recent survey of causal learning in GBRs [17] traces four open challenges to this gap: out-of-distribution generalisation, dynamic preferences, fairness and interpretability. All four arise because message passing propagates correlations of the observational graph rather than causal preference structure. The same survey lists the lack of theoretical guarantees, scalability and privacy as open directions.

Two families of fixes exist. *Loss-level* debiasing (IPS, DR) reweights the training objective [12, 13, 15] but leaves the propagation operator biased. *Propagation-level* debiasing reweights edges by inverse propensities during neighbour aggregation [6, 8]. We show that the second family is itself biased for multi-hop propagation. The root cause is elementary. The edge estimate $W_e = O_e Y_e / p_e$ is unbiased for the potential outcome Y_e , but a walk that traverses e twice contributes W_e^2 , and $\mathbb{E}[W_e^2] = Y_e / p_e \neq Y_e$. Walks with repeated edges are not rare corner cases. They include every back-and-forth step $u \rightarrow j \rightarrow u$ that makes LightGCN self-reinforcing. Their bias is largest precisely where the propensity is smallest.

Contributions.

1. We identify the *repeated-walk bias* of inverse-propensity and DR graph propagation. We give its exact form and a lower bound that is unbounded as propensities vanish (Theorem 2).
2. We propose DRUP. Using $Y_e^2 = Y_e$, DRUP replaces each repeated edge by its idempotent unbiased counterpart. The resulting 3-hop operator is exactly unbiased and edge-wise doubly robust for propagation over the counterfactual full-exposure graph (Theorem 1). It has the cost of one item Gram matrix, needs no training, and factorises into a public item operator plus per-user local corrections.
3. We prove variance, clipping-bias and concentration bounds (Theorem 3).
4. We show that the same structure yields provable *FAT+P* properties: causal item fairness via exposure invariance (Theorem 4); exact attributions and optimal counterfactual explanations (Proposition 5, Theorem 6); certified robustness to injected profiles (Theorem 7); and joint differential privacy (Theorem 8).
5. We evaluate DRUP on Coat [13], whose test set is a random sample, and on KuaiRec [3], whose test matrix is fully observed. We compare it with trained and training-free baselines and verify each guarantee: by Monte Carlo, by a real-data intervention on exposure, by brute force, and against realised attacks.

2 Related work

Causal and debiased recommendation. Inverse propensity scoring [13] and doubly robust learning [15] are the standard tools for MNAR feedback. Causal embeddings disentangle interest from conformity [18]. The survey of [17] organises causal GBRs by challenge. None of these methods analyses the bias of the *propagation operator* beyond one hop.

Debiased message passing. NAVIP [8] applies inverse-propensity weights with Laplacian normalisation inside LightGCN aggregation. DPAA [6] learns popularity-aware aggregation weights. Both are unbiased for a single aggregation step at most. Theorem 2 shows that stacking such steps reintroduces a bias of order $1/p$.

Linear graph filters. LightGCN is a polynomial filter of the normalised adjacency, and closed-form filters are competitive with it [14]. DRUP is a closed-form filter on a debiased operator. Its linearity is what makes exact explanations, certificates and DP releases possible.

Explanations, robustness and privacy. GNNExplainer [16] and CF-GNNExplainer [9] approximate explanations by optimising on a surrogate. PORE [7] certifies recommenders against poisoning by ensembling. Differentially private recommenders release noisy item covariances [11]. DRUP gives exact explanations, closed-form certificates and a joint-DP guarantee [5] for one and the same scoring rule.

3 Problem setting

Users $u \in [m]$, items $i \in [n]$ and pairs $e = (u, i)$ carry a binary potential outcome Y_e and an exposure $O_e \sim \text{Bernoulli}(p_e)$. The log reveals O and $O \odot Y$.

Assumption 1 (Unconfoundedness). *Given the propensities, the exposures O_e are mutually independent and independent of Y .*

Assumption 2 (Fixed nuisances). *The propensity estimate $\hat{p}$, the imputation $\hat{Y} \in [0, 1]^{m \times n}$ and the edge weights C do not depend on O . This holds with cross-fitting or with nuisances estimated on a vetted historical log. We use $C_{ui} = d_u^{-\alpha} d_i^{-(1-\alpha)}$ with degrees taken from $\hat{Y}$.*

With $\tilde{Y} = C \odot Y$, the target is propagation over the full-exposure graph:

$$F^* = a\tilde{Y} + bT^*, \quad T^* = \tilde{Y}\tilde{Y}^\top \tilde{Y}. \quad (1)$$

This is the linear LightGCN / graph-filter score one would compute if every item had been shown to every user.

4 DRUP

Edge estimates. With clipped propensities $\bar{p} = \max(\hat{p}, \tau)$,

$$W_e^{\text{IPS}} = \frac{O_e Y_e}{\bar{p}_e}, \quad W_e^{\text{DR}} = \hat{Y}_e + \frac{O_e(Y_e - \hat{Y}_e)}{\bar{p}_e}. \quad (2)$$

Both have mean Y_e when $\bar{p} = p$.

Idempotent walk correction. Let $\tilde{W} = C \odot W$ and $\Delta = \tilde{W} \odot \tilde{W} - C \odot C \odot W$, with row sums r and column sums κ . The corrected 3-hop operator is

$$\hat{T} = \tilde{W}\tilde{W}^\top \tilde{W} - \tilde{W} \odot (r\mathbf{1}^\top - \Delta) - \tilde{W} \odot (\mathbf{1}\kappa^\top - \Delta) - (\tilde{W}^{\odot 3} - C^{\odot 3} \odot W). \quad (3)$$

The three corrections handle walks $u-j-v-i$ with $v = u$, with $j = i$, and with both. In each case every repeated factor $\tilde{W}_e^k$ is replaced by $C_e^k W_e$. The DRUP score is $s = a\tilde{W} + b\hat{T}$ with global constants a, b .

Public-operator form. Define the diagonal-corrected item Gram matrix $G = \tilde{W}^\top \tilde{W} - \text{diag}(\tilde{W}^\top \tilde{W}) + \text{diag}(\mathbf{1}^\top (C \odot C \odot W))$. Then

$$\hat{T}_u = \tilde{w}_u G - \tilde{w}_u \odot (r_u \mathbf{1}^\top - 2\delta_u) - (\tilde{w}_u^{\odot 3} - c_u^{\odot 3} \odot w_u), \quad (4)$$

where $\delta_u = \tilde{w}_u^{\odot 2} - c_u^{\odot 2} \odot w_u$. Equation (4) matches (3) to machine precision. G is the only object shared across users (Algorithm 1).

Algorithm 1 DRUP (training-free)

Require: $\log(O, O \odot Y)$; hyper-parameters $\tau, \alpha, \lambda, b/a$

- 1: fit propensities $\hat{p}$ (logistic exposure model) and imputation $\hat{Y}$ (IPS-weighted additive ridge)
 - 2: $\bar{p} \leftarrow \max(\hat{p}, \tau)$; $W \leftarrow \hat{Y} + O \odot (Y - \hat{Y})/\bar{p}$
 - 3: $C \leftarrow d_u^{-\alpha} d_i^{-(1-\alpha)}$ with d from $\hat{Y}$
 - 4: $G \leftarrow \tilde{W}^\top \tilde{W} - \text{diag}(\tilde{W}^\top \tilde{W}) + \text{diag}(\mathbf{1}^\top (C \odot C \odot W))$ $\triangleright$ public operator
 - 5: **for** each user u (locally) **do**
 - 6: $\hat{T}_u \leftarrow \text{Eq. (4)}$; $s_u \leftarrow a \tilde{w}_u + b \hat{T}_u$; rank unexposed items by s_u
 - 7: **end for**
-

Complexity. One $n \times n$ Gram product and $O(mn)$ reductions, i.e., the cost of the uncorrected operator. On KuaiRec (7,176 users, 10,728 items, 10.3M exposures) one configuration takes under a minute on four CPU cores.

5 Theory

Full proofs are in Appendix A. Let $\varepsilon_e = |Y_e - \hat{Y}_e|$.

Theorem 1 (Edge-wise double robustness). *Under Assumptions 1–2, let $\delta_e = (p_e/\bar{p}_e - 1)(Y_e - \hat{Y}_e)$. Then $\mathbb{E}[\hat{T}_{ui}]$ equals T_{ui}^* with every Y_e replaced by $Y_e + \delta_e$. In particular, if every edge has a correct propensity or a correct imputation, then $\mathbb{E}[s] = F^*$.*

Theorem 2 (Repeated-walk bias of uncorrected propagation). *With correct propensities and a candidate (u, i) with $O_{ui} = 0$, the uncorrected DR propagation $\tilde{W}\tilde{W}^\top \tilde{W}$ has conditional bias*

$$C_{ui}\hat{Y}_{ui}(R_u + K_i) - C_{ui}^3\hat{Y}_{ui}(1 - \hat{Y}_{ui}^2), \quad K_i = \sum_{v \neq u} C_{vi}^2 \frac{1 - p_{vi}}{p_{vi}} \varepsilon_{vi}^2,$$

and R_u is the analogous sum over the user’s own edges. The within-user ranking bias $C_{ui}\hat{Y}_{ui}K_i$ satisfies $K_i \geq (1/\max_v p_{vi} - 1) \sum_{v \neq u} C_{vi}^2 \varepsilon_{vi}^2$. It therefore diverges as the item’s propensities vanish, and it is $\Omega(1/\tau)$ under clipping. DRUP’s bias is zero.

Theorem 3 (Variance, clipping and concentration). *Let $\Lambda_e(u, i)$ be the C -weighted mass of the walks from u to i through edge e , and let $B = \max(1, 1/\tau)$. Then (a) $\text{Var}(W_e) \leq \varepsilon_e^2/\tau$; (b) $\text{Var}(\hat{T}_{ui}) \leq \frac{(1+1/\tau)^2}{\tau} \sum_e \varepsilon_e^2 \Lambda_e(u, i)^2$; (c) $|\mathbb{E}\hat{T}_{ui} - T_{ui}^*| \leq 3\bar{\delta}(1 + \bar{\delta})^2 \sum_{j,v} C_{uj}C_{vj}C_{vi}$ with $\bar{\delta} = \max_e(1 - p_e/\tau) + \varepsilon_e$; (d) $\Pr(|\hat{T}_{ui} - \mathbb{E}\hat{T}_{ui}| \geq t) \leq 2 \exp(-2t^2/\sum_e c_e^2)$ with $c_e = \varepsilon_e B^2 \Lambda_e(u, i)/\tau$.*

The variance scales with the imputation error ε^2 , not with Y^2 as for IPS, and the clip τ trades variance against a clipping bias that vanishes when $\hat{Y}$ is accurate.

Definition 1 (Exposure elasticity). For an intervention $do(p_{\cdot i} \leftarrow \pi p_{\cdot i})$ on item i ’s exposure mechanism, $\eta_i = \partial \log \mathbb{E}[s_{ui}]/\partial \log \pi$.

Theorem 4 (Exposure invariance: causal item fairness). *Under Assumptions 1–2 and $\pi p \geq \tau$, DRUP has $\eta_i = 0$, so items with equal potential outcomes receive equal expected scores whatever their exposure history. Logged-graph propagation has $\eta_i \geq 1$ (popularity amplification). Uncorrected DR propagation has $\eta_i < 0$ whenever $\hat{Y}_{ui}K_i > 0$ (over-correction).*

Proposition 5 (Affine in the user’s own data). *Fix u and a candidate i with $O_{ui} = 0$, and let $G^{(-u)}$ be G without u ’s own contribution. Then $s_{ui} = \text{const} + \sum_{j \neq i} (c_{uj} G_{ji}^{(-u)} + c_{ui} w_{ui} c_{uj}^2) w_{uj}$. Hence un-logging interaction j changes s_{ui} by exactly $\phi_j = a_j(w_{uj} - \hat{Y}_{uj})$. Removal effects add up over any subset, the ϕ_j are the exact Shapley values, and completeness holds exactly.*

Theorem 6 (Optimal minimal counterfactual explanation). *For candidates i, k with $s_{ui} > s_{uk}$, sorting $g_j = \phi_j^{(i)} - \phi_j^{(k)}$ in decreasing order and removing interactions until the cumulative sum exceeds the margin yields a minimum-cardinality set whose removal makes k outrank i . If the positive g_j do not sum above the margin, no such set exists. The cost is $O(n + d_u \log d_u)$.*

Theorem 7 (Certified robustness to injected profiles). *With frozen nuisances, one injected user v changes u ’s 3-hop score of item k by $\Delta_k = x_k(D + a_k c_{vk}) - a_k x_k^2$, where $a = \tilde{w}_u$, $x = c_v \odot w_v$ and $D = \langle a, x \rangle$. If v logs at most L interactions with arbitrary items, labels and propensities $\geq \tau$, then D and x_k lie in closed-form intervals. Δ_k is linear in D and quadratic in x_k , so its extremes $[\underline{\Delta}_k, \overline{\Delta}_k]$ over this box are exact. Item t provably stays out of u ’s top- K against any F such users if $s_{ut} + F \overline{\Delta}_t$ is below the $(K+1)$ -th largest value of $s_{uk} + F \underline{\Delta}_k$. With unclipped IPS, the influence of a single interaction is unbounded.*

Theorem 8 (Joint differential privacy). *Scale each user row so that $\|\tilde{w}_v\|_2 \leq R$ and $\|c_v^{\odot 2} \odot w_v\|_2 \leq R^2$. Releasing G plus symmetric Gaussian noise with $\sigma = 2R^2 \sqrt{2 \ln(1.25/\delta)}/\epsilon$ is (ϵ, δ) -DP under adding or removing one user. Scoring by (4) uses only the release and the user’s own row, so the recommender is (ϵ, δ) -jointly DP [5]. Post-processing such as low-rank denoising is free.*

6 Experiments

6.1 Setup

Data. *Coat* [13] has 290 users and 300 items. Training consists of 6,960 self-selected (MNAR) ratings. Testing uses 16 ratings per user on random items (MAR), and we rank only items the user never rated. The dataset ships propensities; we also fit our own exposure model, with the same conclusions. *KuaiRec* [3] provides the big matrix (7,176 users, 10,728 items, 10.3M logged views) as the MNAR log and a fully observed small matrix (1,411 users, 3,327 items, 99.6% dense) as the test set. We rank all never-logged small-matrix items. Positives are ratings ≥ 4 (Coat) and watch ratio ≥ 2 (KuaiRec).

Protocol. Hyper-parameters are chosen on a validation part of the unbiased data: 30% of each user’s test items for Coat, 30% of the test users for KuaiRec. We report the held-out remainder, averaged over 10 (Coat) or 5 (KuaiRec) random splits; trained baselines use 10 and 3 splits. The baselines are popularity, imputation only, and MF / IPS-MF / DR-MF / LightGCN / NAVIP-LightGCN / DR-LightGCN trained with early stopping. The training-free propagation family shares one scoring form and differs only in its operator: logged graph (linear LightGCN), IPS adjacency (NAVIP-style), IPS with walk correction (WC), DR adjacency, and DRUP.

6.2 Monte-Carlo verification

Table 1 uses a synthetic MNAR model with known propensities, 20,000 replications for bias and 3,000 for elasticity. The uncorrected IPS and DR operators have relative biases of 2.4 and 1.9. With the walk correction, bias disappears: the fraction of entries with $|z| > 3$ matches the 0.27% expected by chance, and RMSE drops about fivefold. Elasticity follows Theorem 4: +1.78 for the logged graph, −0.19 for DR adjacency, and ≈ 0 for DRUP.

Table 1: Monte-Carlo check of Theorems 1, 2 and 4.

Estimator	rel. —bias—	frac. $ z > 3$	rel. RMSE	exposure elasticity η
Obs	0.992	1.0000	1.02	+1.777
IPS	2.403	0.4042	21.01	+0.007
IPS+WC	0.028	0.0025	4.93	+0.007
DR	1.903	0.6292	20.94	-0.187
DRUP	0.020	0.0008	3.97	+0.005

Table 2: Accuracy on unbiased test data (mean $\pm$ std over splits). Best in bold, second underlined.

Method	metrics (Coat: nDCG@5, Recall@5, nDCG@10; KuaiRec: nDCG@20, Recall@20, nDCG@50)		
<i>Coat</i>			
Popularity	0.5367±0.0095	0.6441±0.0093	0.6599±0.0086
Imputation only	0.5662±0.0095	0.6736±0.0120	0.6787±0.0071
MF	0.4619±0.0224	0.5666±0.0213	0.6056±0.0178
IPS-MF	0.4569±0.0234	0.5579±0.0241	0.6049±0.0178
DR-MF	0.5357±0.0141	0.6473±0.0162	0.6575±0.0113
LightGCN	0.5077±0.0173	0.6036±0.0216	0.6403±0.0109
NAVIP-LightGCN	0.4994±0.0199	0.5979±0.0246	0.6329±0.0129
DR-LightGCN	0.5560±0.0241	0.6565±0.0254	0.6725±0.0165
Linear LightGCN (logged graph)	0.5571±0.0100	0.6461±0.0094	0.6745±0.0076
IPS adjacency (NAVIP-style)	0.5438±0.0186	0.6403±0.0152	0.6641±0.0138
IPS adjacency + WC	0.5438±0.0186	0.6403±0.0152	0.6641±0.0138
DR adjacency	<u>0.5586±0.0124</u>	<u>0.6728±0.0143</u>	0.6737±0.0088
DRUP (ours)	0.5526±0.0134	0.6637±0.0157	0.6720±0.0108
<i>KuaiRec</i>			
Popularity	0.5766±0.0023	0.1656±0.0016	0.4610±0.0019
Imputation only	0.6215±0.0014	0.1867±0.0020	0.5112±0.0013
Linear LightGCN (logged graph)	0.6036±0.0028	0.1815±0.0016	0.4985±0.0024
IPS adjacency (NAVIP-style)	0.5954±0.0022	0.1829±0.0015	0.4892±0.0017
IPS adjacency + WC	0.5954±0.0022	0.1829±0.0015	0.4892±0.0017
DR adjacency	<u>0.6318±0.0017</u>	<u>0.1927±0.0023</u>	<u>0.5144±0.0019</u>
DRUP (ours)	0.6326±0.0018	0.1929±0.0025	0.5147±0.0015

6.3 Accuracy

Table 2 reports accuracy. DRUP needs no training, yet it is on par with the best trained baselines on Coat and is the most accurate method on KuaiRec. On these data the walk correction changes accuracy only marginally relative to DR adjacency; the correction matters for bias, fairness and certification, not for top- K accuracy. As predicted by Theorem 2, IPS+WC and IPS coincide on unexposed candidates at three hops, because the factor $W_{ui} = 0$ hides the IPS bias.

6.4 Fairness

Table 3 tests Theorem 4 on real data. We thin the logged exposures of a random half of the items with probability 1/2, which is an intervention with a known propensity change. Logged-graph and (clipped) IPS propagation demote the thinned items, i.e., they punish under-exposure. DRUP’s rank shift is statistically indistinguishable from zero. The NDCG gap between user groups is also smaller for the DR-based operators. *Limitation:* exposure invariance is a causal guarantee, not a distributional one. When true quality is concentrated, DRUP’s top- K lists are also concentrated, and its Gini and PRU are higher than those of the logged graph on Coat. The accuracy–concentration frontier in Table 4 makes this trade-off explicit. Combining DRUP with an exposure-constrained re-ranker is left for future work.

Table 3: Fairness. ECB is the partial correlation between an item’s mean rank and its log exposure, controlling for its true like rate. The user gap is the nDCG gap between user groups (gender on Coat, activity on KuaiRec). The last column is the rank shift of items whose exposures were thinned by a real $do(p \leftarrow p/2)$; 0 means exposure invariance.

Data	Method	nDCG	ECB↓	PRU↓	Gini↓	user gap↓	rank shift under $do(p/2)$
Coat	Obs	0.4917	0.769	0.394	0.340	0.1123	-0.0585±0.0073
Coat	IPS	0.4864	0.765	0.392	0.337	0.1189	-0.0594±0.0066
Coat	DR	0.4770	0.744	0.763	0.655	0.0795	-0.0027±0.0122
Coat	DRUP	0.4803	0.739	0.758	0.655	0.0793	+0.0006±0.0110

Table 4: Best nDCG reachable by each operator under a cap on the Gini coefficient of top- K exposure (over the whole hyper-parameter grid).

Data	Method	Gini≤0.3	≤0.4	≤0.5	≤0.6	any
Coat	Obs	–	0.4917	0.4917	0.4917	0.4917
Coat	IPS	–	0.4864	0.4864	0.4864	0.4864
Coat	DR	–	0.4090	0.4487	0.4733	0.4853
Coat	DRUP	–	0.3786	0.4416	0.4848	0.4902

6.5 Transparency

Attributions recomputed against full re-propagation are exact to machine precision, both for completeness and for additivity over random subsets (Table 5). Minimal counterfactual explanations are small, and where none exists the explanation is itself informative. On Coat the top recommendation of most users cannot be overturned by removing any subset of their own interactions: it is driven by population-level evidence, which DRUP can certify.

6.6 Accountability: certified robustness

Table 6 shows that clipping plus imputation-based degrees make DRUP the only operator with non-trivial certificates. The certificate was never violated by a realised attack, and brute force over all small profiles confirms that the bound is valid and reaches 62–67% of the true worst case. Empirically, DR adjacency with attacker-controlled degrees can be harder to push at intermediate budgets, because the fake user’s large DR weights inflate its own degree. That protection is uncertifiable, since the attacker can choose a profile that avoids it.

6.7 Privacy

Table 7 shows the privacy–utility curve. Coat has only 290 users, so meaningful privacy costs accuracy there. KuaiRec, with 25 times more users, supports much smaller ϵ .

7 Discussion and limitations

DRUP relies on unconfoundedness and on nuisances that are fixed or cross-fitted, the standard assumptions of IPS/DR estimation. The exact correction is derived for up to three hops (and for the item Gram operator). Higher orders need the same idempotent rule, but the number of repetition patterns grows. The fairness guarantee concerns causal exposure invariance, not parity of

Table 5: Exact explanations (Prop. 5, Thm. 6) on 200 random test users.

Data	max rel. error	mean min. CF size	median	users without CF
Coat	1.3e-15	2.36	2	0.82

Table 6: Injection attack: F fake users log a low-popularity target item and the most popular items. Left: realised hit rate of the target in real users’ top- K . Right: fraction of real users for whom Theorem 7 certifies that the target cannot enter the top- K against *any* F profiles of that size. No certificate was violated.

hit rate of target in top- K ↓							certified fraction ↑					
<i>Coat</i> (K=5, profile size 21)												
Method	$F=0$	$F=1$	$F=5$	$F=10$	$F=20$	$F=50$	$F=0$	$F=1$	$F=5$	$F=10$	$F=20$	$F=50$
Obs	0.004	0.004	0.009	0.108	0.444	0.661	0.996	0.000	0.000	0.000	0.000	0.000
IPS	0.004	0.004	0.020	0.201	0.502	0.651	0.995	0.000	0.000	0.000	0.000	0.000
DR	0.000	0.000	0.000	0.000	0.000	0.655	1.000	0.000	0.000	0.000	0.000	0.000
DRUP	0.000	0.000	0.000	0.000	0.382	0.973	1.000	1.000	0.211	0.000	0.000	0.000

outcomes. The privacy analysis treats the low-dimensional nuisances as public. Beyond accuracy, DRUP turns a graph recommender into an object whose bias, fairness, explanations, robustness and privacy can be stated and checked.

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Table 7: nDCG of jointly DP DRUP ($\delta = 10^{-5}$). The denoising rank is chosen on validation data.

Data	$\epsilon=0.5$	1	2	4	8	16	∞ (clipped)	non-private / Pop
Coat	0.4094	0.4118	0.4182	0.4651	0.5068	0.5301	0.5398	0.5326 / 0.5226

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A Proofs

Theorem 1. After the correction every walk contributes $\prod_{e \in \text{distinct}} C_e^{k_e} W_e$. The W_e are independent (Assumption 1, fixed nuisances), so $\mathbb{E} \prod W_e = \prod \mathbb{E} W_e$ with $\mathbb{E} W_e = \hat{Y}_e + p_e(Y_e - \hat{Y}_e)/\bar{p}_e = Y_e + \delta_e$. The target monomial is $\prod C_e^{k_e} Y_e^{k_e} = \prod C_e^{k_e} Y_e$ because $Y_e \in \{0, 1\}$. $\square$

Theorem 2. The naive operator contains, for $v = u$, $j \neq i$, the terms $\tilde{W}_{uj}^2 \tilde{W}_{ui}$; for $j = i$, $v \neq u$, the terms $\tilde{W}_{ui} \tilde{W}_{vi}^2$; and $\tilde{W}_{ui}^3$. Conditionally on $O_{ui} = 0$ we have $W_{ui} = \hat{Y}_{ui}$. By independence, $\mathbb{E}[W_e^2] = Y_e + \text{Var}(W_e)$ with $\text{Var}(W_e) = \frac{1-p_e}{p_e}(Y_e - \hat{Y}_e)^2$, while the corrected replacement has mean Y_e . Summing the excesses gives the formula. The lower bound uses $(1-p)/p \geq 1/\max p - 1$. $\square$

Theorem 3. (a) $\text{Var}(W_e) = p_e(1-p_e)\varepsilon_e^2/\bar{p}_e^2$. If $p_e \geq \tau$ this is at most $(1/\tau - 1)\varepsilon_e^2$; otherwise it is at most $p_e\varepsilon_e^2/\tau^2 < \varepsilon_e^2/\tau$. (b) $\hat{T}_{ui}$ is multilinear in independent variables and therefore affine in each W_e . The Efron–Stein inequality [2] gives $\text{Var}(\hat{T}) \leq \sum_e \mathbb{E}[(\partial_e \hat{T})^2] \text{Var}(W_e)$. The derivative $\partial_e \hat{T}$ is a sum, over the walks through e , of C -weights times at most two other independent W_f with $\|W_f\|_2^2 \leq 1 + 1/\tau$. Minkowski’s inequality then bounds $\|\partial_e \hat{T}\|_2 \leq (1 + 1/\tau)\Lambda_e$. (c) By Theorem 1, the bias is at most $\sum |\kappa_{\text{walk}}| |\prod (Y_f + |\delta_f|) - \prod Y_f| \leq ((1 + \bar{\delta})^3 - 1)M_{ui}$, and $|\delta_e| = (1 - p_e/\tau)_+ \varepsilon_e$. (d) Changing O_e moves W_e by at most ε_e/τ , and $|\partial_e \hat{T}| \leq B^2 \Lambda_e$. Apply McDiarmid’s inequality [10]. $\square$

Theorem 4. DRUP: by Theorem 1, $\mathbb{E}[s] = F^*$, which does not depend on π . Logged graph: $\mathbb{E}[O_e Y_e] = p_e Y_e$, so each 3-hop monomial containing $k \geq 1$ edges incident to i scales as π^k . Uncorrected DR: by Theorem 2, $\mathbb{E}[s] = F^* + C_{ui} \hat{Y}_{ui} K_i(\pi) + \dots$, and $K_i(\pi)$ is strictly decreasing in π . $\square$

Proposition 5 and Theorem 6. Substitute $G = G^{(-u)} + \tilde{w}_u^\top \tilde{w}_u - \text{diag}(\tilde{w}_u^2) + \text{diag}(c_u^2 w_u)$ into (4). The terms $\tilde{w}_{ui} \sum_j \tilde{w}_{uj}^2$ cancel against the local correction, leaving $\tilde{w}_{ui} \sum_j c_{uj}^2 w_{uj}$, and every remaining term involving w_{uj} ($j \neq i$) is linear. Additive games have Shapley values equal to their individual effects. For Theorem 6, the margin after removing S is $s_{ui} - s_{uk} - \sum_{j \in S} g_j$, so minimising $|S|$ under $\sum_S g_j > \text{margin}$ selects the largest g_j . $\square$

Theorem 7. Adding row v adds contrib_v to G and leaves u ’s local terms unchanged. Then $\Delta = \tilde{w}_u \text{contrib}_v = (a \cdot x)x - a \odot x^2 + a \odot c_v \odot x$. Logged DR entries satisfy $w \in [\hat{Y} - \hat{Y}/\tau, \hat{Y} + (1 - \hat{Y})/\tau]$ for any label and any $p \geq \tau$, and unlogged ones equal $\hat{Y}$. D is maximised (minimised) by assigning the L slots to the largest positive (negative) deviations. For fixed x_k the function is linear in D , so it attains its extremes at the ends of the D interval. For fixed D it is a quadratic in x_k , whose extremes over an interval lie at the ends or at the clamped vertex. Effects add over fake users. For unclipped IPS, $|W_e| = 1/p_e$ is unbounded. $\square$

Theorem 8. $\|\text{contrib}_v\|_F \leq \|\tilde{w}_v \tilde{w}_v^\top - \text{diag}(\tilde{w}_v^2)\|_F + \|c_v^2 w_v\|_2 \leq R^2 + R^2$, so the ℓ_2 -sensitivity of the upper triangle is at most $2R^2$, and the Gaussian mechanism [1] applies. Each user’s output is a function of the DP release and their own data, which gives joint DP by the billboard lemma [5]. $\square$